INCH-POUND

MIL-DTL-17/52C
w/AMENDMENT1
9 April 2025
SUPERSEDING
MIL-DTL-17/52C
10 April 2015

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS, M17/52-00001, M17/52-RG119, UNARMORED, M17/52-RG120, ARMORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

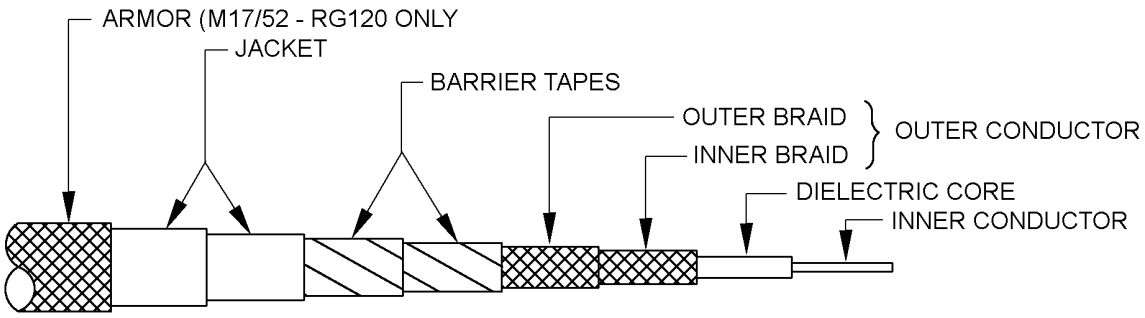


FIGURE 1. Configuration.

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TABLE I. Description.

Components	Construction details
Inner conductor	Solid, bare, copper wire. Diameter: .1019 inch ± .0020.
Dielectric core	Type F-1: Solid, extruded PTFE Diameter: .332 inch ± .005.
Outer conductor	Double braid of bare, copper wire. 1/ Diameter: .414 inch maximum.
Inner braid	Gauge: AWG No. 33 Coverage: 97.7% nominal Carriers: 24 Ends: 10 Picks/inch: 5.4% ± 10
Outer braid	Gauge: AWG No. 34 Coverage: 91.1% nominal Carriers: 24 Ends: 8 Picks/inch: 10.6% ± 10
Barrier tapes	Type FF-2: Two wraps of PTFE tape, .005 ± .0005 inch thick each, by 1 to 1-1/4 inch wide. Each wrap of PTFE tape is to be applied with a 50% minimum overlap.
Jacket	Type V: Double braid. Diameter: .465 inch ± .010.
Armor (M17/52-RG120)	Single braid of aluminum-alloy wire. Diameter: .525 inch maximum.

1/ The braids shall be silver-plated for part number M17/52-00001.

ENGINEERING INFORMATION:

Continuous working voltage: 4,500 V rms, maximum.
Operating frequency: 1 GHz, maximum.

NOTE: The 1 GHz maximum is recommended due to use of bare copper braids. If 3 GHz is required, a silver plate should be added (M17/52-00001 only).

Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55°C to +200°C.
Weight: 22.8 pounds per 100 feet maximum for M17/52-RG119 and M17/52-00001; 28.6 pounds per 100 feet, maximum for M17/52-RG-120.

Inner conductor properties:
DC resistance (maximum at 20°C): 0.103 ohms per 100 feet.
Elongation: 15 percent, minimum.

Engineering notes: This cable is useful in high temperature applications. (See connector series HN per MIL-DTL-3643).

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REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.
Eccentricity: 10 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.

Aging stability: +230°C ± 5°C.
Cold bend: -55°C ± 2°C.
Dimensional stability: +230°C ± 5°C.
Inner conductor from core: .250 inch, maximum.
Inner conductor from jacket: .312 inch, maximum.

Flammability: Applicable.

Electrical:

Test frequency: 50 MHz to 1 or 3 GHz as applicable.
Spark test: Not applicable.
Voltage withstanding: 12,000 V rms, minimum.
Corona extinction voltage: 6,000 V rms, minimum.
Characteristic impedance: 50 ohms ± 2.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 29.3 pF per foot, maximum.

Marking: applicable

Marking Durability: 125 cycles (250 strokes), 1 pound

Workmanship: applicable

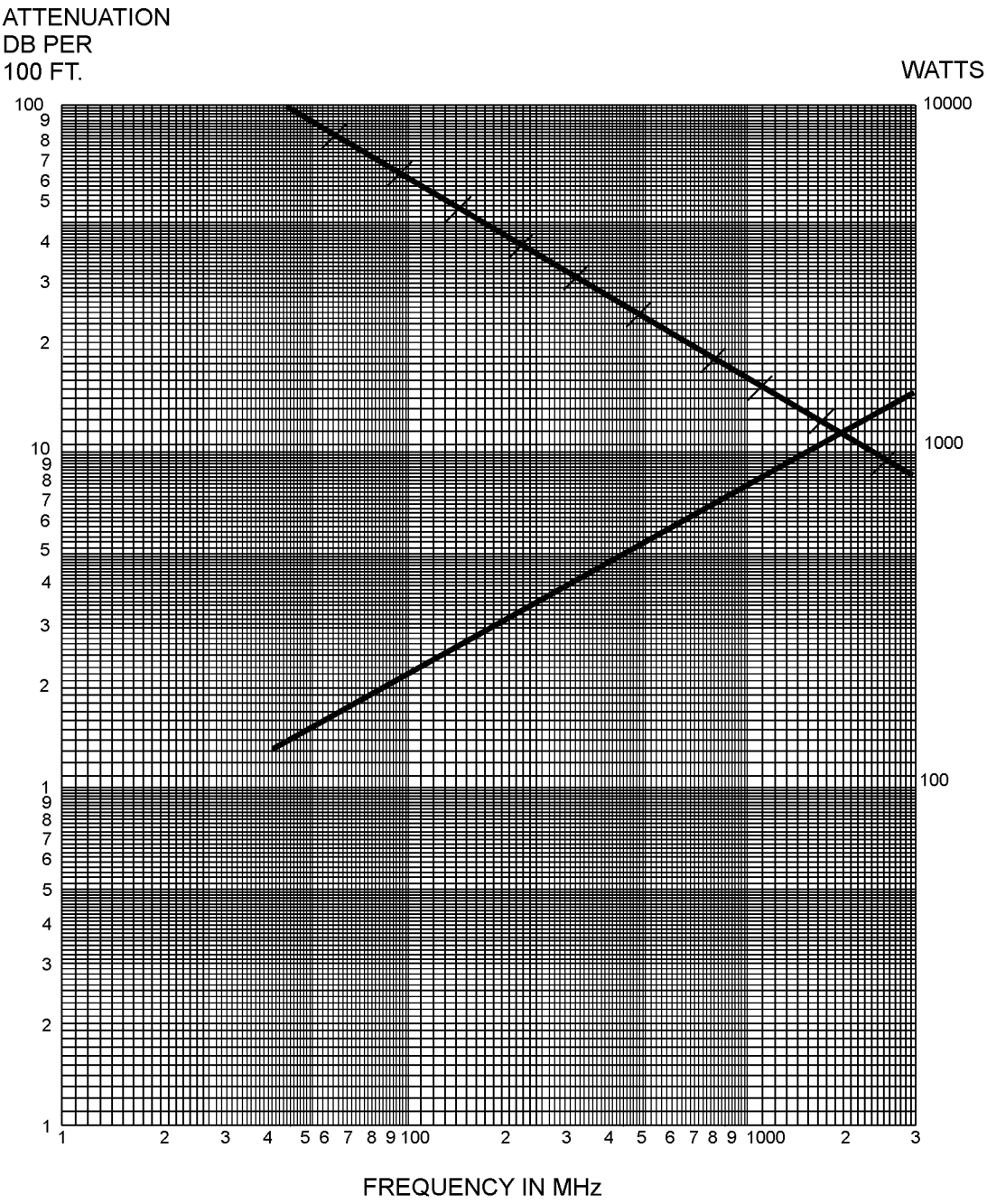
Part or Identifying Number (PIN): See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

PIN	Superseded PIN or type designation
M17/52-RG119	--
M17/52-RG120	RG-120/U
M17/52-00001	--

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Maximum power (at 25°C sea level) ---|---|---|---|---|---

Maximum attenuation ----------

FIGURE 2. Power rating and attenuation.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1
8.7242	.7943	2
5.8480	.7078	3
4.4194	.6310	4
3.5655	.5623	5
3.0095	.5018	6
2.6146	.4467	7
2.3228	.3951	8
2.0909	.3548	9
1.9250	.3162	10
1.7849	.2618	11
1.6709	.2512	12
1.5769	.2239	13
1.4985	.1995	14
1.4326	.1778	15
1.3767	.1585	16
1.3280	.1413	17
1.2880	.1259	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1726	.0794	22
1.1524	.0705	23
1.1347	.0611	24
1.1192	.0562	25
1.1055	.0501	26
1.0935	.0447	27
1.0829	.0398	28
1.0736	.0355	29
1.0633	.0318	30

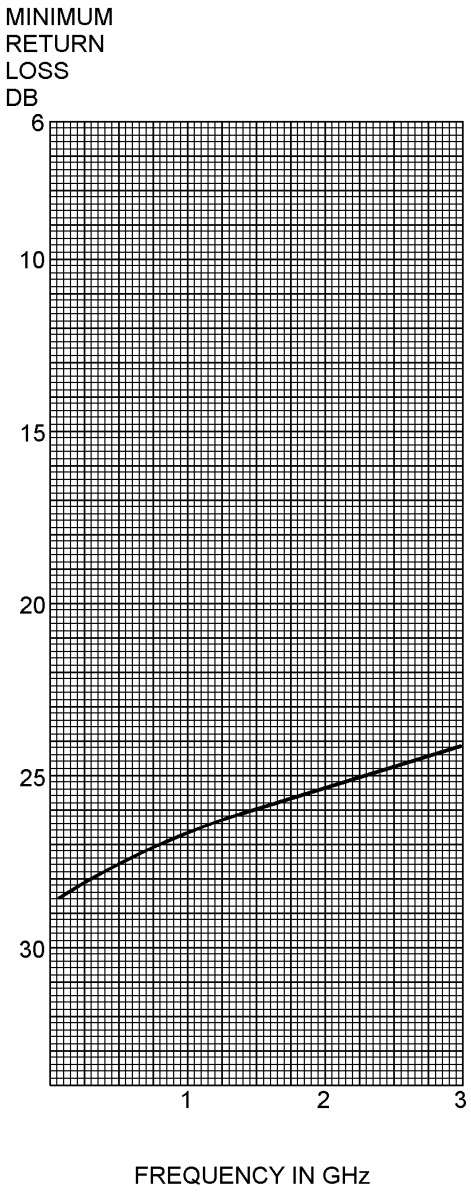


FIGURE 3. Structural return loss.

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TABLE I. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				

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TABLE I. Qualification and Group A, B, and C Table – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	3.7.22	4.8.23	1	Per Table VIII	C
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion	N/A				
	Smoke index	N/A				
	Toxicity index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-DTL-3643

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2025-009)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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